WG.48.2: Δ and Ξ as Operators

Proposal for the Z Standard, by Wolfgang Grieskamp (wg@cs.tu-berlin.de), November 11, 1998, (v3)

What

Δ and Ξ are introduced as expression operators. An implicit definition of a schema named ΔS (ΞS) is not supported.

Why

The current handling of implicit definition of ΔS and ΞS schemas introduces an irregularity into Standard Z which complicates the language definition and implementation. This irregularity is in particular problematic in combination with paragraph reordering (see WG.48.3).

From a methodological point of view, the possibility of attaching non-standard meanings to the phrase ΔS can be criticised. If a special meaning of ΔS is required, it seems to be better style to document this meaning in a descriptive name for this schema.

Since in Standard Z schema references have been generalised to expressions, it would be orthogonal to treat Δ and Ξ as operators on expressions. This allows e.g. specifications such as

$$
\begin{array}{l}
D == [x, y : \mathbb{Z} \mid x > y] \\
\Delta_Dx == [\Delta D; \Xi(D \mathbin{\mathsf{fn}} x)] \\
Op == [\Delta_Dx \mid x' = x + 1]
\end{array}
$$

How

Updates to the WD 1.5

- Characters: introduce Δ and Ξ as symbols
- Lexis: introduce Δ and Ξ as keywords
- Concrete and annotated syntax: introduce Δ and Ξ as unary operators. Assign them a priority above schema decoration.
- Transformations: reduce Δ and Ξ expressions.

Compatibility Specifications which make use of the redefinition of ΔS can be easily ported by a global textual replace of ΔS with e.g. Δ_S. Specifications which don't make use of the redefinition don't need to be changed – a single lexem Δ followed by a schema name S is the application of the operator Δ to the name S.

Variants Instead of treating Δ and Ξ as symbols, it would be also possible to keep them being letters. Distinguishing between the name ΔS and the operation $\Delta\ S$ has to be indicated by whitespace, as with decoration. Type checkers will detect that a name ΔS (without space) is not defined.